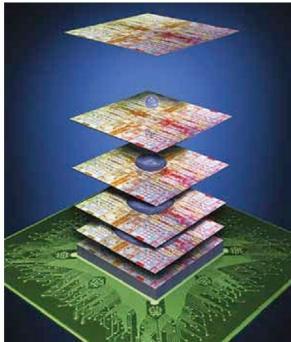


cal plane assignment of standard cell/macros to minimise wire length and via count. Clock tree synthesis and routing must try to minimise congestion using TSVs, and it is critical that clocking is balanced across planes as well across each die."

According to Sawicki, for 2.5D, DRC and LVS verification challenges are relatively modest. Dies are developed/verified using the existing design and verification flows. Then the die interfaces, usually silicon interposers, are designed and verified using a similar flow with some enhancements to handle multi-die pin-out naming checks. Parasitic extraction tools are being enhanced to handle silicon interposers, front and backside metal layers, and inter-die coupling, which require new models for these elements. For full 3D (stacked die with TSVs), there are lots of new challenges such as thermal modelling (including the impact of IR drop), power network integrity, stress modelling, and modelling of the parasitic effects of TSVs interacting with each other and with transistors in active die areas.

In the test area, the biggest requirement is providing known-good-die (KGD) to the 2.5D or 3D assembly process in order to ensure a profitable yield. The term 'KGD' means more comprehensive testing needs to be used during wafer-level testing. For 3D (stacked die) in particular, the test access infrastructure for routing test data and results to and from any of the dies within the vertical stack is a critical need.

This is a new requirement because all but the bottom dies have connections only to their immediate neighbours and no direct access to package pins. A dedicated access infrastructure must be incorporated within each die of a vertical stack in order to route test data and results up and down the stack. Since the bare die may come from different sources, standardisation of this infrastructure will be needed to ensure



3D IC stacking (Image courtesy: www.i-micronews.com)

proper interoperability between all dies. The IEEE P1838 working group is currently developing such a standard with direct involvement of Mentor.

Sawicki shares: "Multi-die scenarios also create the need for system representations that can accurately represent new elements in a consistent manner, and allow rapid prototyping and optimisation across multiple dies. The EDA solutions for 2.5D (interposed based) are well in hand and do not pose a technology barrier. Full 3D (chip stacking with TSVs) for homogeneous designs (e.g., processors or other logic systems split across multiple dies) is another matter. That will definitely require significant EDA development and some re-architecting of the design tools. These tools are now in development and are expected to be available when the market demand materialises."

Collaborative effort required

Some kind of collaborative approach is essential to harness 2.5D/3D-IC technologies successfully.

Ahuja explains: "Collaboration and building a well-defined ecosystem are crucial for successful design and usage of 3D ICs. To support 3D IC for EDA

tools, there are additional components which need a deeper level of understanding. Optimising system cost with the shortest possible turnaround time is a major challenge which is not possible without an integrated approach to 3D-IC design. Since many 3D stacks combine digital and analogue/RF circuitry, a strong analogue/mixed-signal capability plus a robust IC/package co-design capability and PCB layout options are critical for providing a 'complete' 3D-IC realisation methodology.

"3D-IC design should be a shared effort among system architects, package designers, IC designers of various dies, PCB designers and DFT engineers. That calls for a system that can handle the handshake between different platforms, close collaboration between different design environments, and co-design among groups that have historically worked separately. Handoffs between the same company's designers are difficult enough, and 3D-IC die handoffs could be between different companies or participants in the ecosystem."

According to Ahuja, while EDA tools can help minimise some of these interactions by providing a common platform, the overall design task gets challenging when dies come from different places and are implemented in different environments. A proper handoff point must be agreed upon by the industry to make it easier for IC designers to exchange design data. Therefore collaboration within the ecosystem becomes crucial.

Standards development activities are going on at the Silicon Integration Initiative (Si2), SEMI and Sematech. The IEEE P1838 Test Working Group is working on 3D-IC DFT architectures. In the memory domain, there is a lot of activity going on at JEDEC, which has ratified the wide input/output (I/O) memory standard. Micron and Samsung have jointly announced formation of the High density Memory